Combining Stopping and Working Memory Slows Response Inhibition and Relates to Network Integration

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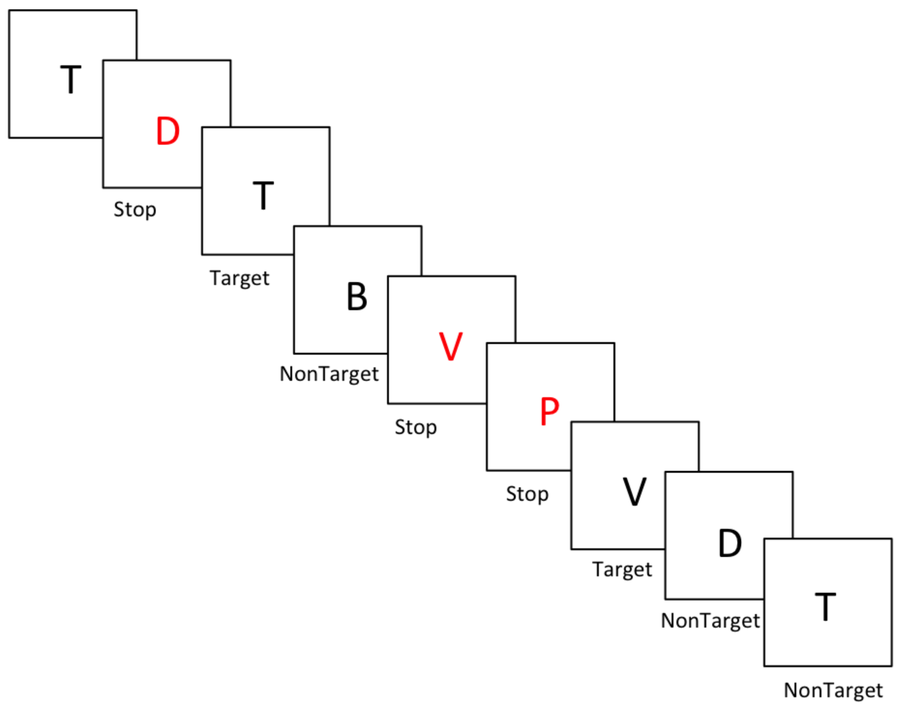
Introduction

Extant taxonomies of cognitive control suggest that working memory and response inhibition are separable[1]. However, controlled behavior requires flexible cooperation between these acts of control.

We interrogated the relationship between working memory (i.e., updating active memory traces and resisting proactive interference from recent memory traces) and response inhibition in a novel combination of the N-back[2] and stop-signal tasks[3], respectively.

We investigated the network-level signatures underlying successful performance in this novel dual task.

Method/Analysis



Task: Example 2-back block in N-BackStop task. If the current letter matched the stimulus presented N trials ago, subjects responded with one button press (target trials, 1/3 of all trials) but otherwise responded with a different button press (nontarget trials, 2/3 of all trials). The letter turned red on 1/3 of trials, which acted as a stop signal, so subjects were instructed to make neither button press. The load (i.e., N) of the N-back task was adjusted based upon error rate on no-stop trials. Subjects were instructed to remember all letters in the sequence even if they turned red.

fMRI: N=29 subjects underwent a 90-minute MRI scanning session (GE 3T scanner) using a multiband acquisition sequence (MB=8; TR=680ms; voxel size=2.2mm; 64 axial slices). All subjects were run through fMRI quality assurance (github.com/poldracklab/mriqc)[4] and preprocessing (github.com/poldracklab/fmriprep)[5] pipelines developed by the Stanford University Center for Reproducible Neuroscience.

Figure 1: N=20 agreed to return for a follow-up behavioral study that included four tasks: a retest (FollowUp), a 1-back stopping task (1-BackStop), a no working memory load stopping task (NoLoadStop), and an adaptive N-back task with no stop signals (N-BackNoStop). We present their stop-signal reaction times (SSRTs) across conditions.

Figure 2: We computed regional time-varying functional connectivity matrices using the Multiplication of Temporal Derivatives (MTD)[6]. We then classified each region according to its profile of within- (W_T; module-degree Z-score) and between- (B_T; participation coefficient) module connectivity. The joint histogram (100x100) of these two measures is used to estimate the extent of integration and segregation within the functional connectome in each temporal window. We display the interaction of a 2x2 ANOVA comparing trialtype (N-back go vs. N-back stop) and response (Correct vs. Error) (FDR-corrected at .05).

Figure 3: We computed a second time-varying functional connectivity measure, the jackknife correlation[7], and we computed the volatility of this measure[8], which is the average change in the connectivity matrix from timepoint to timepoint. We ask whether higher volatility may relate to effective performance in this task environment.

Results

Figure 1. Response inhibition (SSRT) slowed as working memory load increased (indexed by N-back load) (ANOVA p < .01)

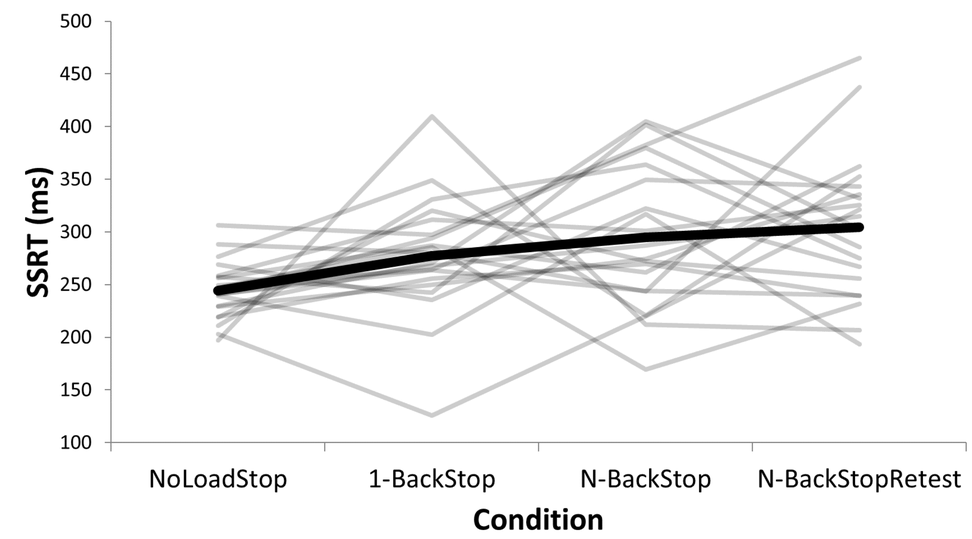


Fig 2. Global network integration (B_T) increased during correct stopping trials (hot colors) compared to failed go trials (cool colors)

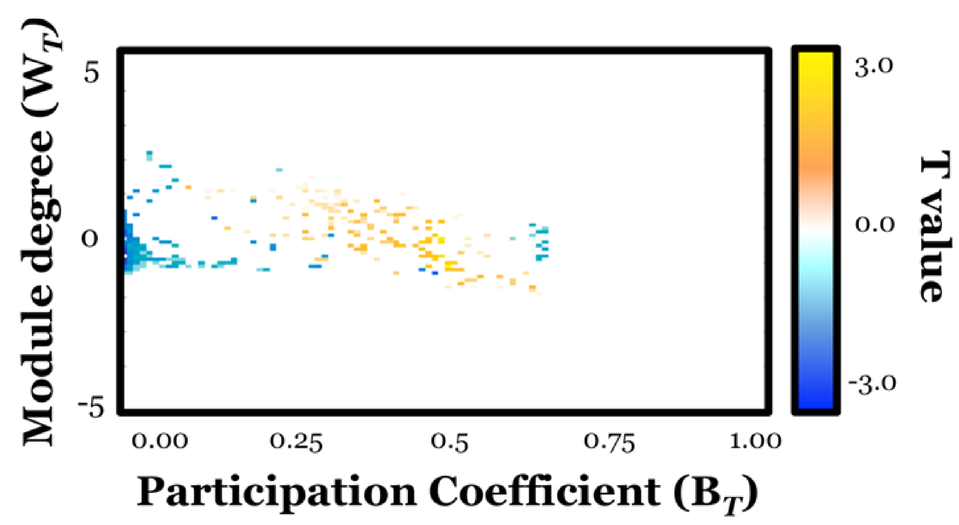
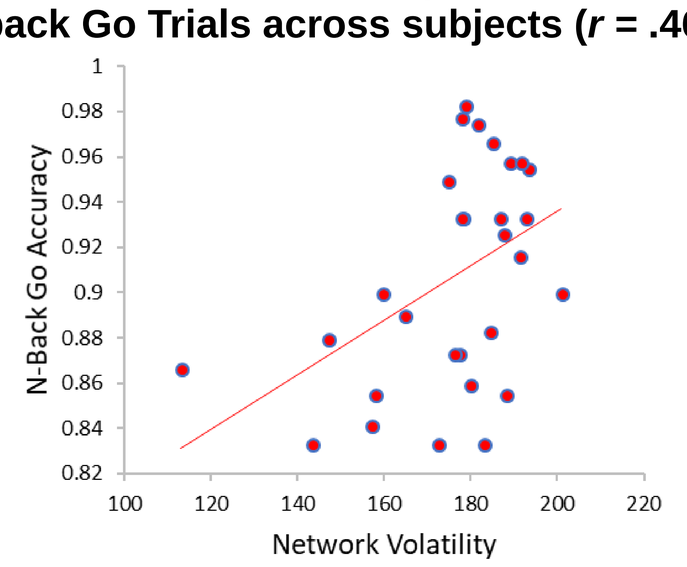


Figure 3. Global network volatility related to better accuracy on 2-back Go Trials across subjects (*r* = .46, *p* = .01)



Conclusions

SSRT slowed as N-back load increased. This challenges existing taxonomies by suggesting that response inhibition and working memory are not separable.

Global functional network integration related to successful trial-by-trial performance on N-back stop-signal trials. This suggests that cooperative recruitment across functional networks is important for cooperative recruitment of different acts of control.

Subjects with less stable networks were more accurate on 2-back go trials, suggesting that being able to quickly and/or frequently reconfigure network patterns may be beneficial for dual tasking.

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